Constrained damping for looped two-stream reasoners: summary of results

November 2026

1 Question

The October summary [1] compared two exactly invertible two-stream layers, Håkon's *kick* coupling (the conformal layer of [2]) and our *relax* coupling, as looped reasoners. Kick lost almost all of its correct answers when iterated beyond the training horizon; relax was stable on multi-hop composition [3]. Here we train tiny looped models from scratch on two synthetic tasks and ask

1. what both couplings do far beyond the number of iterations they were trained with, and
2. whether constraining their damping makes them *stop* once they have the answer.

2 Architecture: what goes wrong, and two constraints

As before, the state has two streams L, H; attention updates L and reads H, the MLP does the converse. The input embedding e is added back at every iteration, and the prediction is read from H:

$$\textbf{kick:}\quad L \leftarrow e^{-\gamma}\big(L + \mathrm{Attn}(\mathrm{LN}(H+e))\big), \qquad H \leftarrow e^{-\alpha}H + \mathrm{MLP}(\mathrm{LN}(L+e)),$$

$$\begin{aligned}\textbf{relax:}\quad L &\leftarrow e^{-\gamma}L + (1-e^{-\gamma})\big(H + \mathrm{Attn}(\mathrm{LN}(H+e))\big),\\ H &\leftarrow e^{-\alpha}H + (1-e^{-\alpha})\big(L + \mathrm{MLP}(\mathrm{LN}(L+e))\big).\end{aligned}$$

Training draws the number of iterations uniformly from 2–8 (3–8 on 3-hop) and takes the loss after the last one.

Why unconstrained loops drift. Switch off attention and MLP. Relax then leaves every state with $L = H$ unchanged: that direction is neither damped nor pulled back, and the attention and MLP outputs add up along it. The state therefore grows *linearly* with the number of iterations whenever these outputs keep pointing, on average, in a fixed direction. The architecture allows this but does not force it. In the trained relax models it does happen: out to 2048 iterations, the running average of the outputs tends to a fixed nonzero vector, and the current output lines up with it (cosine 1.00). As the state grows, e becomes a vanishing part of what attention and the MLP see, so the loop forgets its input.

Kick with rates near zero is bounded in principle, but only on a long time scale. With $\gamma = \alpha = 0.005$, as used before, it keeps 99.5% of its state per iteration. A perturbation therefore takes about 200 iterations to decay, and the state can grow to about 200 times the size of the attention and MLP outputs before it is held back. The trained model does exactly this: its state grows for about 1000 iterations and then levels off.

Two constraints. Both keep exact invertibility and the constant volume factor $e^{-d\sum(\gamma+\alpha)}$.

- **Relax with common-mode damping $\delta > 0$.** Scale the cross-stream terms by $c = e^{-\delta} < 1$:

$$L \leftarrow e^{-\gamma} L + (1 - e^{-\gamma})\big(c\, H + \mathrm{Attn}(\mathrm{LN}(H + e))\big),$$
$$H \leftarrow e^{-\alpha} H + (1 - e^{-\alpha})\big(c\, L + \mathrm{MLP}(\mathrm{LN}(L + e))\big).$$

  Now every direction is damped. For $\gamma = \alpha = 3$, with attention and MLP off, the slowest direction shrinks by a factor 0.97, 0.89 and 0.64 per iteration for $\delta = 1/64$, $1/16$ and $1/4$ (for $\delta = 0$ the factor is 1).

- **Kick with positive rates.** The rates $\gamma, \alpha = 0.05 + \mathrm{softplus}(\theta)$ are learned, starting from 1, so the state is damped by at least $e^{-0.05} \approx 0.95$ per iteration.

With every direction damped and bounded attention and MLP outputs, the state stays bounded, and the input cannot be forgotten.

Training beyond the horizon (carry). Following Equilibrium Reasoners [4], in half of the training steps we first run $r_0 \in \{1, \ldots, 64\}$ iterations (uniform) without gradient, then the usual 2–8 with gradient. The loss thus sees states up to 72 iterations deep, while gradients still pass through at most 8; this costs on average 16 extra forward-only iterations per training step.

3 Results

Setup. ListOps [5] with MIN and MAX, nesting depth at most 3 (6000 training steps), and 3-hop pointer chasing: 8 pairs $a \to f(a)$, then 8 queries q with answers $f(f(f(q)))$ (8000 steps). Models have 32 dimensions per stream, 4 heads, one layer per iteration and about 18k parameters. We report accuracy on 2000 held-out sequences after up to 128 iterations.

	ListOps					3-hop pointer chasing				
model	n	acc@8	acc@128	O [%]	$\Delta\hat{y}$ [%]	n	acc@8	acc@128	O [%]	$\Delta\hat{y}$ [%]
standard block	3	0.918 ±0.015	0.143 ±0.048	85.4 ±5.3	85.1 ±5.0	1	1.000	0.990	1.0	1.0
relax	3	0.917 ±0.018	0.262 ±0.112	72.4 ±12.5	73.3 ±11.1	3	0.990 ±0.014	0.935 ±0.079	5.8 ±7.0	5.8 ±7.0
kick, $\gamma = \alpha = 0.005$	1	0.843	0.193	78.5	80.6	1	0.999	0.155	84.5	84.5
kick, positive rates	1	0.893	0.892	0.7	2.1	1	0.913	0.567	38.8	37.0
relax, $\delta = 1/4$	1	0.866	0.567	36.0	42.2	1	0.991	0.947	4.5	4.5
kick, positive rates + carry	3	0.918 ±0.019	0.916 ±0.021	0.5 ±0.3	1.3 ±0.4	3	0.962 ±0.052	0.946 ±0.071	2.4 ±3.0	2.7 ±3.5
relax, $\delta = 1/4$ + carry	3	0.944 ±0.005	0.946 ±0.005	0.4 ±0.0	1.4 ±0.4	3	0.998 ±0.001	0.909 ±0.024	9.0 ±2.3	9.0 ±2.3

Table 1: Main results. Test accuracy after 8 and 128 iterations; O: of the answers correct at 8 iterations, the percentage that is wrong at 128 (lower is better); $\Delta\hat{y}$: percentage of predictions that differ between 8 and 128 iterations. Mean ± s.d. over n seeds. "Standard block": the standard transformer block, looped. Groups: no constraint on the damping; constraint; constraint and carry.

4 Takeaways

- **Without constraints, looped models overthink, relax included.** On ListOps the looped standard block and relax lose 85% and 72% of their correct answers by 128 iterations (three seeds); kick with rates 0.005 loses 79%. The state grows linearly and the input's weight

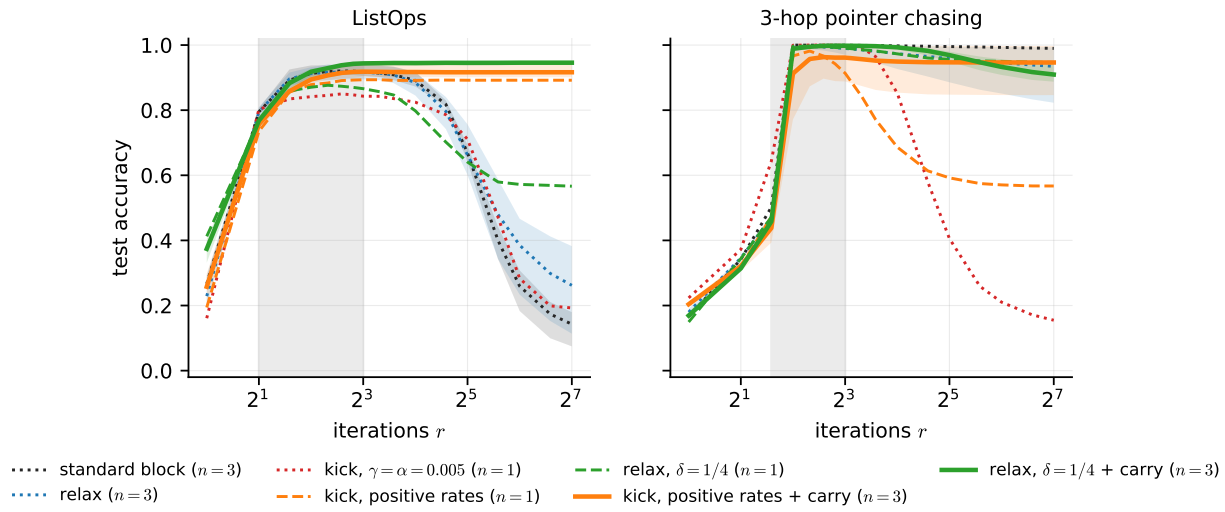


Figure 1: Test accuracy against the number of iterations; grey band: iterations used in training. Dotted: no constraint; dashed: constraint; solid: constraint and carry. Shaded: range over seeds.

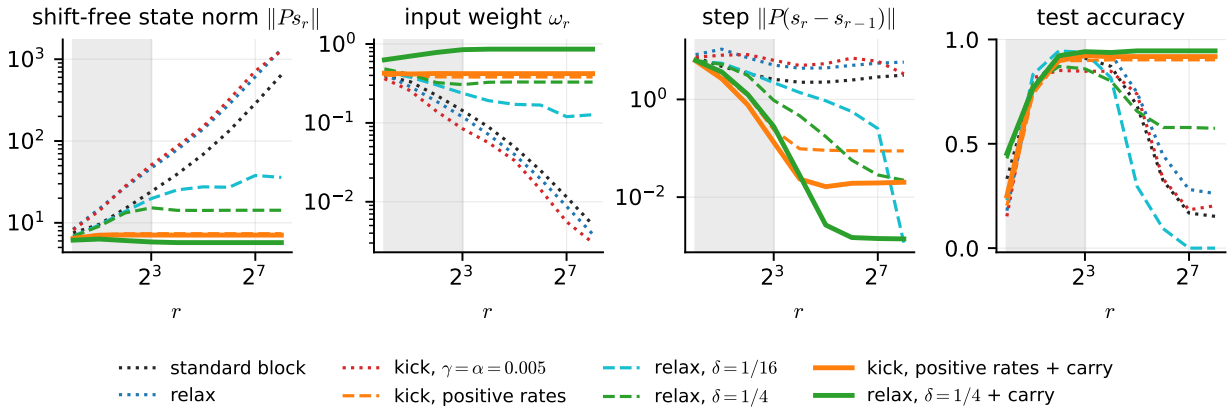


Figure 2: Trained ListOps models (one seed) iterated far beyond training. Left to right: size of the state, weight of the input e inside $\mathrm{LN}(H + e)$, change of the state per iteration, and accuracy. Per-token means, which LayerNorm cannot see, are removed first. Without constraints the state grows linearly and the input's weight falls like $1/r$. With constraints both stay flat.

	ListOps				3-hop pointer chasing			
model (seed 1238)	acc@8	acc@128	$\|\Delta s\|_{128}$	ρ_{128}	acc@8	acc@128	$\|\Delta s\|_{128}$	ρ_{128}
standard block	0.910	0.179	2.9	1.00	1.000	0.990	1.6	1.00
relax	0.938	0.290	5.6	1.00	0.999	0.988	2.4	1.00
kick, $\gamma = \alpha = 0.005$	0.843	0.193	6	1.00	0.999	0.155	1.2	1.00
kick, free learned rates	0.844	0.833	7e+10	1.20	1.000	0.016	–	–
kick, positive rates	0.893	0.892	0.11	0.69	0.913	0.567	0.022	0.89
relax, $\delta = 1/64$	0.904	0.069	1.2	0.98	0.999	0.996	0.24	0.98
relax, $\delta = 1/16$	0.926	0.000	0.26	0.94	0.999	0.997	0.014	0.93
relax, $\delta = 1/4$	0.866	0.567	0.03	0.94	0.991	0.947	0.0026	0.92
relax, learned δ and rates	0.900	0.107	0.048	0.98	0.999	0.991	0.067	0.96
relax, post-LN injection	0.816	0.474	6.5	1.00	0.978	0.845	1.6	1.00
standard block, post-LN injection	0.799	0.511	5	1.00	0.997	0.978	1.8	1.00
relax, $\delta = 1/16$, post-LN injection	0.803	0.230	0.52	0.98	0.939	0.272	0.14	0.99
relax + carry	0.801	0.773	0.96	1.00	0.749	0.581	0.2	1.00
standard block + carry	0.787	0.744	0.57	1.00	–	–	–	–
relax, $\delta = 1/16$ + carry	0.783	0.746	0.076	0.98	0.665	0.398	0.024	0.99
kick, positive rates, zero-init	–	–	–	–	0.889	0.607	0.041	0.87
kick, positive rates + carry	0.907	0.902	0.015	0.63	0.999	0.997	0.05	0.75
relax, $\delta = 1/4$ + carry	0.937	0.939	0.0015	0.83	0.999	0.942	0.0021	0.92

Table 2: All configurations, one seed. $\|\Delta s\|_{128}$: change of the state at iteration 128 (lower: more settled). ρ_{128}: local contraction factor per iteration at iteration 128, measured on the trained model (below 1: the state is pulled towards a fixed point; 1: it drifts). "Post-LN injection" adds the input after the LayerNorm, $\mathrm{Attn}(\mathrm{LN}(H) + e)$, as a control for forgetting the input.

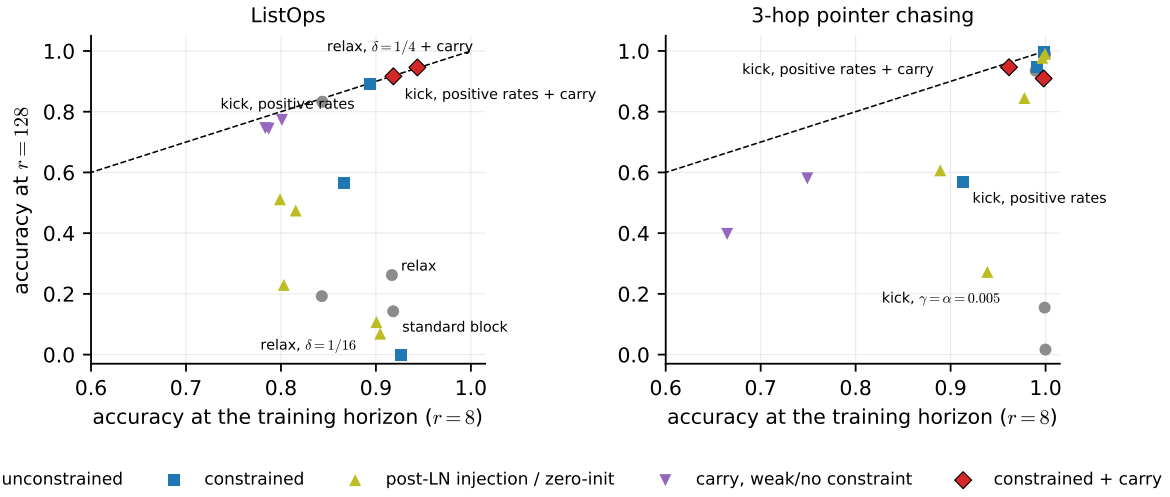


Figure 3: Accuracy at 8 iterations against accuracy at 128, for every configuration in Table 2. On the diagonal the two accuracies are equal; individual answers can still change (Table 1 reports how many do).

falls like $1/r$ (Fig. 2). Kick with rates 0.005 levels off only after about 1000 iterations, on mostly wrong answers. On 3-hop the standard block and relax mostly survive (relax loses 6% on average, but 16% in one seed), presumably because the answer forms early. So relax's stability on multi-hop composition in the October summary does not carry over to every task.

- **Kick is not structurally unstable; its rates were too small.** With positive learned rates (it keeps 25–45% of each stream per iteration), kick keeps its ListOps answers (0.893 → 0.892) with a bounded state. (Kick with free learned rates also keeps them, but its state diverges and it collapses on 3-hop.) This revises the October takeaway: the failures there came from rates near zero or rates learned negative. What matters is how strong the damping is compared with the number of iterations, not just that it is positive.

- **Settling is not the same as settling on the answer.** Every constrained model settles: after 1024 iterations 79–100% of the test sequences sit at a fixed point, and the rest on short cycles. But relax with $\delta = 1/16$ settles on wrong answers on ListOps (0.926 → 0.000) while being near perfect on 3-hop. Nothing in training ties the fixed point to the answer.

- **Constraint plus carry gives accuracy and stopping on ListOps, but only kick does so on both tasks** (Table 1, Fig. 3). On ListOps, over three seeds:

  - relax with $\delta = 1/4$ and carry goes 0.944 → 0.946, against 0.917 → 0.262 for relax;
  - kick with positive rates and carry goes 0.918 → 0.916;
  - in both, only about 1.4% of the predictions change between 8 and 128 iterations.

  On 3-hop, kick with carry mostly holds (0.962 → 0.946; one seed loses 7%). Relax with $\delta = 1/4$ and carry is accurate at 8 iterations (0.998) but loses 9% of its correct answers by 128, while settling: again a fixed point that is not tied to the answer. The carry without a strong constraint *hurts*: the carried states have drifted, which costs 12–33 points at 8 iterations (one seed).

- **Open.** The models are tiny and the tasks synthetic, and most ablations use one seed. Length generalisation (longer problems than in training [6]) is untested. Strong damping also makes the exact inverse numerically useless over many iterations, so memory-free backpropagation through the loop is not an option; the carry does not need it.

References

[1] Conformal relaxation coupling for looped reasoning models: Summary of results. Internal note, October 2026.

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[3] Harsh Kohli, Srinivasan Parthasarathy, Huan Sun, and Yuekun Yao. Loop, think, & generalize: Implicit reasoning in recurrent-depth transformers. In *Conference on Language Modeling (COLM)*, 2026. arXiv:2604.07822.

[4] Benhao Huang, Zhengyang Geng, and Zico Kolter. Equilibrium reasoners: Learning attractors enables scalable reasoning, 2026. arXiv:2605.21488.

[5] Nikita Nangia and Samuel R. Bowman. ListOps: A diagnostic dataset for latent tree learning. In *Proceedings of the NAACL 2018 Student Research Workshop*, 2018.

[6] Jia Liang, Xi Jin, and Liangming Pan. Beyond the training horizon: Mechanisms and limits of length generalization in looped transformers, 2026. arXiv:2609.33144.